

Uses of Moore-Penrose Pseudoinverse

The Constrained Motion web application uses Moore-Penrose pseudoinverses to compute machine dynamics; however, MPI calculation is buried deep in the code. To get familiar with underlying concept of MPI, one can consider more mundane uses.

I pretend to be processing measurement data in two scenarios dealing with two elementary circuits. We can set voltages ($V_1, V_2 \dots$) and measure current I_{sum} with the hope determine the unknown resistances (conductances, which are reciprocals to resistances). Labels $V_1, V_2, V_3, V_{\text{meas}}, R_1, R_2, R_3$ at the component tops are identifiers for components; labels $V_1, V_2, V_3, 0, g_1, g_2, g_3$ at the bottom of components are the nominals (values) of these components. Sorry for V_1, V_2, V_3 playing two roles in schematic drawings, a convention familiar to electrical engineers.

The first circuit sums up currents from three branches; assume that for some reason, you could take only two measurements with different voltages set for three constant voltage sources V_1, V_2 , and V_3 . The sum of these currents is the current through the measurement voltage source V_{meas} (voltage across this source is zero), the typical trick used to measure currents in simulators.

The second circuit sums up currents from two branches; this time, you can make three measurements with different voltages set for two constant voltage sources V_1 and V_2 . The sum of these currents is the current through the measurement voltage source V_{meas} .

Detailed explanation of circuit component definitions is given in Fig. 2. Resistor conductances are ' g_n ' ($=1/R_n$), branch currents are ' I_n ' ($=V_n * g_n$).

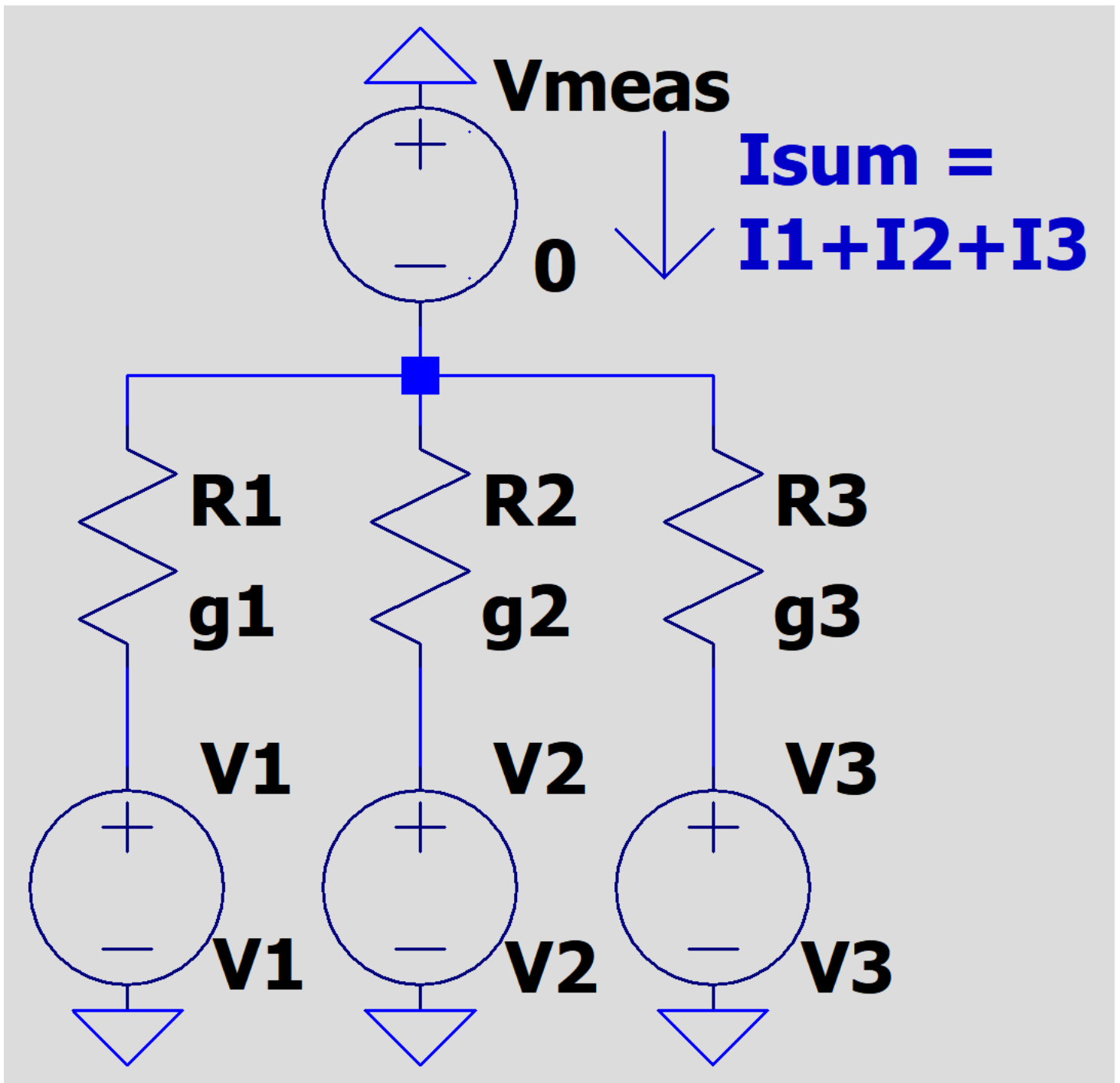


Fig. 1 Three branches, two measurements

Two measurement data sets are available in Scenario 1. The first measurement has voltage source values set as in array $[V1:2, V2:1, V3:0]$, with the current through V_{meas} $I_{sum} = 1$; the second, with $[V1:2, V2:1, V3:0]$ and $I_{sum} = 1$. This gives us a linear system of two equations for three unknowns $[g_1, g_2, g_3]$:

$$\begin{array}{c} / \quad \backslash \\ | \ 2 \ 1 \ 0 \ | \\ | \ 0 \ 1 \ 2 \ | \\ \backslash \quad / \end{array} * \begin{array}{c} / \quad \backslash \\ | \ g_1 \ | \\ | \ g_2 \ | \\ | \ g_3 \ | \end{array} = \begin{array}{c} / \quad \backslash \\ | \ 1 \ | \\ | \ 1 \ | \\ \backslash \quad / \end{array}$$

The convention is to write this system in the form

$$A \cdot g = b$$

As A is not a square matrix, we cannot expect to find a solution in the form $g = A^{-1} \cdot b$. It is where MPI comes to the rescue:

$$g = A^+ \cdot b$$

For linearly independent rows (as is Scenario 1 case), $A \cdot A^*$ is invertible and MPI can be computed as

$$A^+ = A^* \cdot (A \cdot A^*)^{-1}$$

A^* is a transposed matrix with complex conjugated elements. Our matrix A is real, so $A^* = A^T$.

However, $g = A^+ \cdot b$ is not a unique solution for Scenario 1. It is easy to verify that

$$g = A^+ \cdot b + [I - A^+ \cdot A] \cdot w$$

where w is an arbitrary vector $[w_1, w_2, w_3]$, I is identity matrix, also solves the system $A \cdot g = b$. Left multiply $g = A^+ \cdot b + [I - A^+ \cdot A] \cdot w$ by A , and you are back to the source equation:

$$A \cdot g = A \cdot A^+ \cdot b + [A - A \cdot A^+ \cdot A] \cdot w = b + [A - A] \cdot w$$

For completeness, I write down numerical values for above expressions using a programmer-oriented linear notation for matrices.

$$A = [[2, 1, 0], [0, 1, 2]]$$

$$A^* = A^T = [[2, 0], [1, 1], [0, 2]]$$

$$A \cdot A^* = [[5, 1], [1, 5]]$$

$$(A \cdot A^*)^{-1} = [[5/24, -1/24], [-1/24, 5/24]]$$

$$A^+ = A^* \cdot (A \cdot A^*)^{-1} = [[5/12, -1/12], [1/6, 1/6], [-1/12, 5/12]]$$

$$g = A^+ \cdot b + [I - A^+ \cdot A] \cdot w = [1/3, 1/3, 1/3] + [[1/6, -1/3, 1/6], [-1/3, 2/3, -1/3], [1/6, -1/3, 1/6]] \cdot [[w_1], [w_2], [w_3]]$$

We have that $g = [1/3, 1/3, 1/3]$ is a solution, but adding $[[1, -2, 1], [-2, 4, -2], [1, -2, 1]] \cdot [[w_1*6], [w_2*6], [w_3*6]]$ with arbitrary $[[w_1], [w_2], [w_3]]$ also gives a solution. Is there any practical use of expressing solution in this form? Some certainly exists, because although $[[w_1], [w_2], [w_3]]$ can be chosen arbitrarily, the components of vector $[[1, -2, 1], [-2, 4, -2], [1, -2, 1]] \cdot [[w_1*6], [w_2*6], [w_3*6]]$ are linearly dependent. What's more, a physics of your problem can provide additional constraint, such as, for example, non-negativity of conductance (resistance) values.

In linear algebra parlance, Scenario 1 generates an underdetermined system of linear equations.

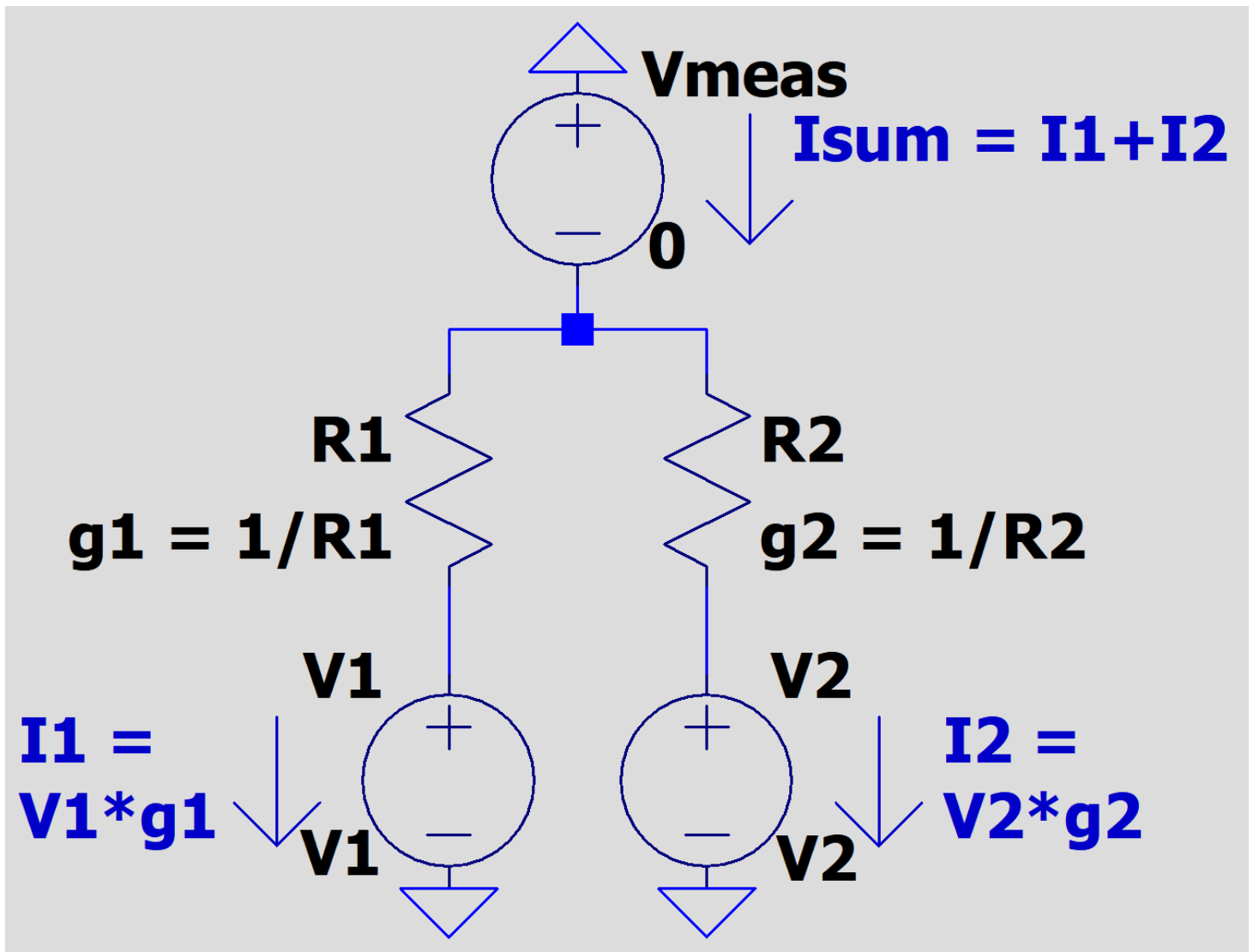


Fig. 2 Two branches, three measurements

Three measurement data sets are available in Scenario 2. The first measurement has voltage source values set as in array $[V1:2, V2:1]$, with the current through V_{meas} $I_{sum} = 2$; the second, with $[V1:1, V2:2]$ and $I_{sum} = 1$ and the third, with $[V1:2, V2:-2]$ and $I_{sum} = -2$. This gives us a linear system of three equations for the two unknowns $[g1, g2]$:

$$\begin{array}{c|c|c|c|c|c|c|c|c|c|} / & \backslash & & / & \backslash & & / & \backslash & & \\ \hline | & 2 & 1 & | & g1 & | & 2 & | & & \\ | & 1 & 2 & | & * & | & g2 & | & = & | 1 | \\ | & 2 & -2 & | & & | & & | & = & | -2 | \\ \hline \backslash & / & & \backslash & / & & \backslash & / & & \end{array}$$

As in the first scenario, we write down this system in the form

$$A \cdot g = b$$

This time we have more measurement data sets than the number of unknown variables. The system is overdetermined, overdetermined systems generally have no exact solution. In idealized experiment, only two equations can be linearly independent and if all the equations happen to be linearly dependent, the g_1 , g_2 values cannot be calculated. In reality, measurement errors make all the equations linearly independent and, under some assumption about error statistics, we can only estimate the most probable g_1 , g_2 values and measurement errors. Sometimes the only information at our disposal is the measured data and then we are using least squares approximation to minimize the sum of the squared 'residuals', the differences between observed values (currents) and the values these have in an ideal measurement setting without measurement errors. Mathematically speaking, we aim to minimize

$$\| A \cdot g - b \|$$

over all measurement instances.

Deciphering $\| \dots \|$ notation for our linear system, we have to minimize

$$(2 \cdot g_1 + g_2 - 2)^2 + (g_1 + 2 \cdot g_2 - 1)^2 + (2 \cdot g_1 - 2 \cdot g_2 + 2)^2$$

that is, to find a vector $[g_1, g_2]$ that minimizes the above expression.

Symbolically, the solution to a 'minimize $\| A \cdot g - b \|$ ' problem is the 'normal equation' so called because this minimization can be interpreted as a geometrical problem. You can learn about the 'normal equation' in textbooks or linear algebra course notes

$$A^T \cdot A \cdot g = A^T \cdot b \text{ (normal equation, } A^T \text{ is the transpose of } A \text{).}$$

Substituting numerical values,

$$A^T \cdot A = \begin{bmatrix} 9 & 0 \\ 2 & 7 \end{bmatrix}$$

$$A^T \cdot b = \begin{bmatrix} 1 & 6 \end{bmatrix}$$

solving equation

$$\text{product}(\begin{bmatrix} 9 & 0 \\ 2 & 7 \end{bmatrix}, [g_1, g_2]) = \begin{bmatrix} 1 & 6 \end{bmatrix}$$

$$\text{we have } g_1 = 1/9, g_2 = 52/63.$$

To solve a general least-square problem and for coding purposes, one can use the Moore Penrose pseudoinverse, likewise it is used to find a general solution for the underdetermined linear system.

Similar to how we do it for an underdetermined system, the expression

$$g = A^+ \cdot b$$

is the solution for minimizes $\|A \cdot g - b\|$ (see prove elsewhere) and this time we should not add the $[I - A^+ \cdot A] \cdot w$ term.

As we have now linearly independent columns, invertible is the $A^+ \cdot A$ product and MPI can be computed as

$$A^+ = (A^+ \cdot A)^{-1} \cdot A^+$$

Substituting values,

$$(A^+ \cdot A) = [[9, 0], [2, 7]]$$

$$(A^+ \cdot A)^{-1} = [[1/9, 0], [-2/63, 1/7]]$$

$$(A^+ \cdot A)^{-1} \cdot A^+ = [[2/9, 1/9, 2/9], [5/63, 16/63, -13/63]]$$

$$g = A^+ \cdot [[2], [1], [-2]] = [1/9, 52/63]$$

The result is identical with that obtained with direct solving of equations.

The sum of all residuals with this solution is

$$(2 \cdot g_1 + g_2 - 2)^2 + (g_1 + 2 \cdot g_2 - 1)^2 + (2 \cdot g_1 - 2 \cdot g_2 + 2)^2 = (-60/63)^2 + (48/63)^2 + (81/63)^2 \approx 3.14$$

A square root of this value is 1.77, much greater than computed conductances. The measurement precision is rather poor.